

Photography problems — Solution

X26 (2026) — English translation

A1

2.00 pts

Find the position and shape of the wire if its length is minimal.

The image of the wire is symmetrical about Y , so the wire lies in the xy plane. Note that each point of the wire gives a point on the screen (if the camera is focused on this point) or a circle (from the likeness of cones).

Let's find the position of the circle and its dimensions for the point with coordinates $(x, y, 0)$. Coordinates of the center of the circle from the similarity of triangles:

$$(Y, Z) = \left(\frac{yl}{x}, 0 \right).$$

For diameter:

$$\frac{d}{D} = \frac{a-l}{a} = 1 - \frac{l}{a}.$$

Substituting value $1/a$ from the formula тонкой линзы.

$$\frac{d}{D} = 1 - l \left(\frac{1}{f} - \frac{1}{x} \right) = 1 - \frac{l(x-f)}{fx}.$$

Let us find зависимость d от Y for $Y > 0$.

By condition $\tan \theta = k$.

$$d = 2Y \sin \theta = \frac{2k}{\sqrt{1+k^2}} \cdot Y = \gamma Y = \gamma \cdot \frac{yl}{x},$$

where $\gamma = \frac{2k}{\sqrt{1+k^2}}$. Substituting это в порауенное ранее expression for d .

$$\frac{d}{D} = \frac{\gamma yl}{Dx} = 1 - \frac{l(x-f)}{fx}$$

Let us express $y(x)$:

$$y = \frac{Dx}{\gamma l} - \frac{D(x-f)}{\gamma f} = \frac{D}{\gamma} - \frac{Dx}{\gamma} \left(\frac{1}{f} - \frac{1}{l} \right) = \frac{D}{\gamma} - \frac{Dx(l-f)}{\gamma fl}.$$

Note that since проволока расположена в области $x > 0$ and image точки проволоки в центре экрана является точечным, то камера сфокусирована на одну from точек проволоки, therefore $l > f$. Note that мы рассматривали случай $a \geq l$, if это не так, то все уравнения останутся верны, но sign у d поменяется. That is for рассмотрения $a < l$ достаточно в порауенных выражениях изменить sign γ . Thus мы we get, что возможные положения точек ограничивающих форму изображения:

$$y = \pm \frac{D(l-f)\sqrt{1+k^2}}{2klf} \left(\frac{lf}{l-f} - x \right).$$

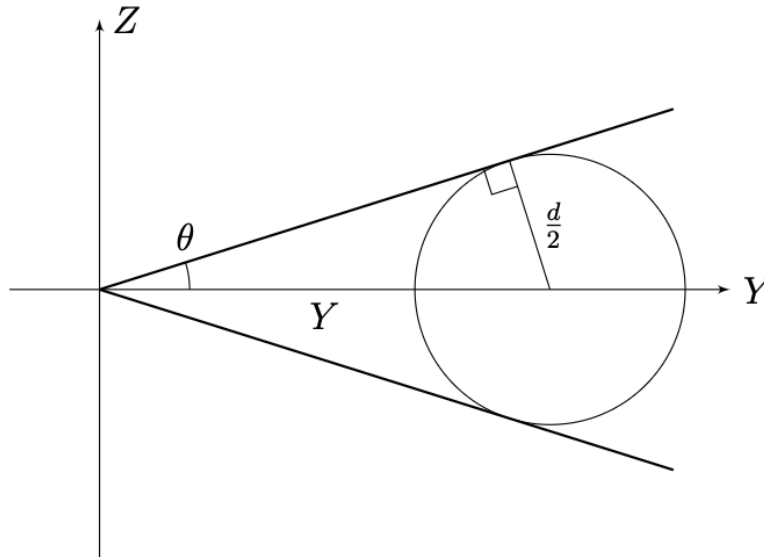
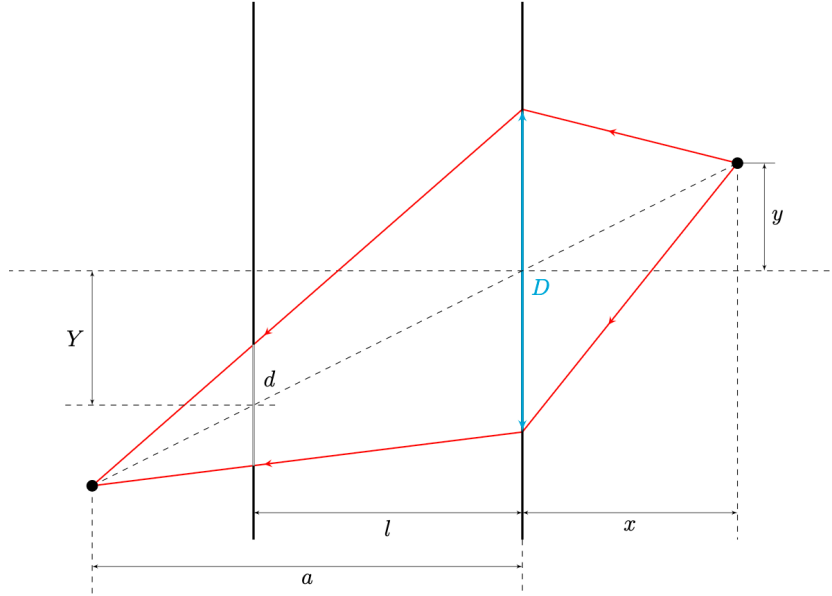
Perhaps the wire contains more sections, the images of which overlap with the images of points on these lines, but then the wire will have a greater length, which contradicts the condition. Therefore, it lies completely on these straight lines.

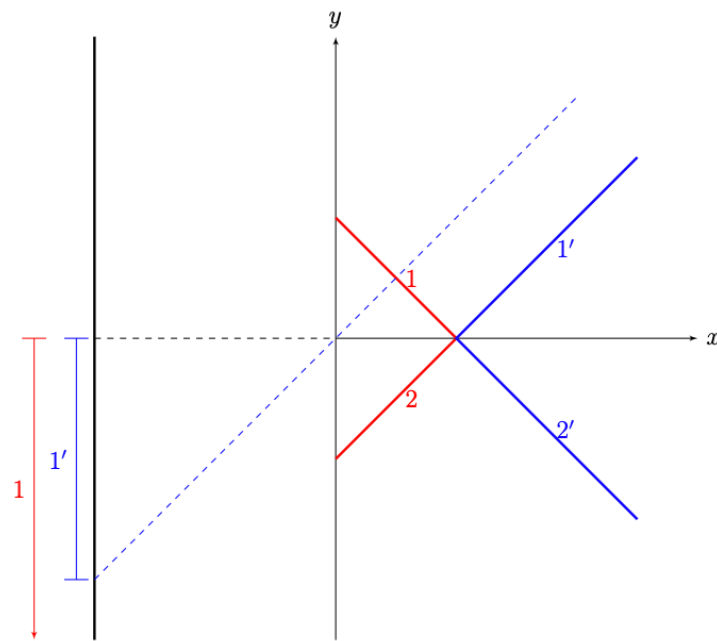
Note that the angular size of the parts of straight lines 1' and 2' is less than 90° , therefore they can give in the image a circle with a limited segment of the positions of the centers; even for a fixed angular size, the length of the wire in sections 1 and 2 will always be less, therefore the wire has the shape of an angle from the segments 1 and 2:

Answer: \textbf{Answer:}

$$y = \pm \frac{D(l-f)\sqrt{1+k^2}}{2klf} \left(\frac{lf}{l-f} - x \right)$$

for/at $x \in \left[0, \frac{lf}{l-f} \right]$.





B1

2.00 pts

Using a compass and straight edge, draw a straight line parallel to the direction of the puck's speed.

Let's understand how an image is formed. Let's enter the coordinates XY , which are attached to the puck and indicate the position of the point on it, and xy , which determine the position in the photograph. Let the speed of the line that the camera films at a particular moment be equal to u ; the center of the puck was photographed at point (x_0, y_0) and at time t_0 .

The angle between the speed of the puck and the x axis will be denoted by α . Let's find the coordinates in the photograph for an arbitrary point on the puck (X, Y) . The moment of photographing this point:

$$t = t_0 + \frac{Y}{u - v \sin \alpha}.$$

Координаты equal:

$$x = x_0 + X + v \cos \alpha (t - t_0) = x_0 + X + \frac{v \cos \alpha}{u - v \sin \alpha} Y,$$

$$y = y_0 + u (t - t_0) = y_0 + \frac{u}{u - v \sin \alpha} Y.$$

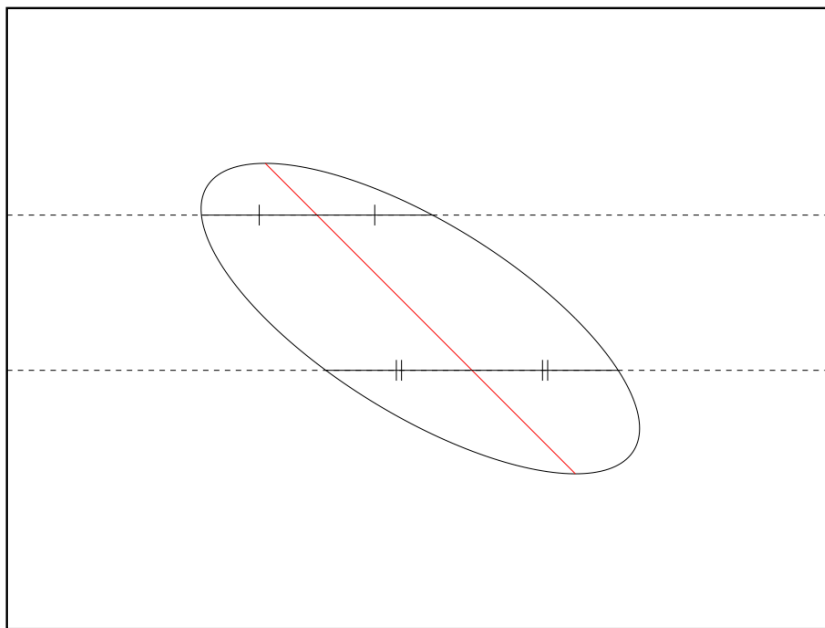
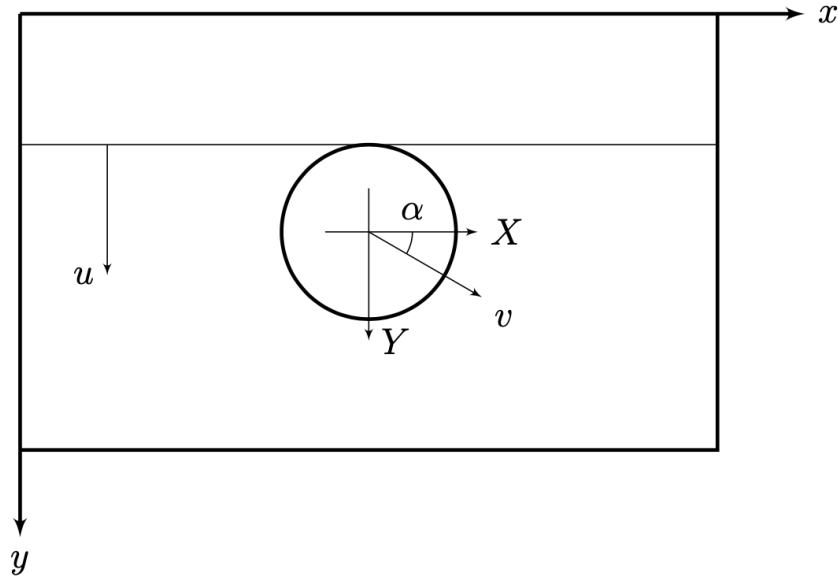
Note that преобразование координат линейно. Значит, прямые на шайбе являются прямыми на изображении. Построим возможные направления скорости. We get image прямой $X = 0$, for этого проведём горизонтальные прямые, image $X = 0$ будет прямой проходящей via/through центры пересечений горизонтальных прямых с imageм шайбы (since for/at фиксированном Y and соответственно y координаты x and X differ only by a constant).

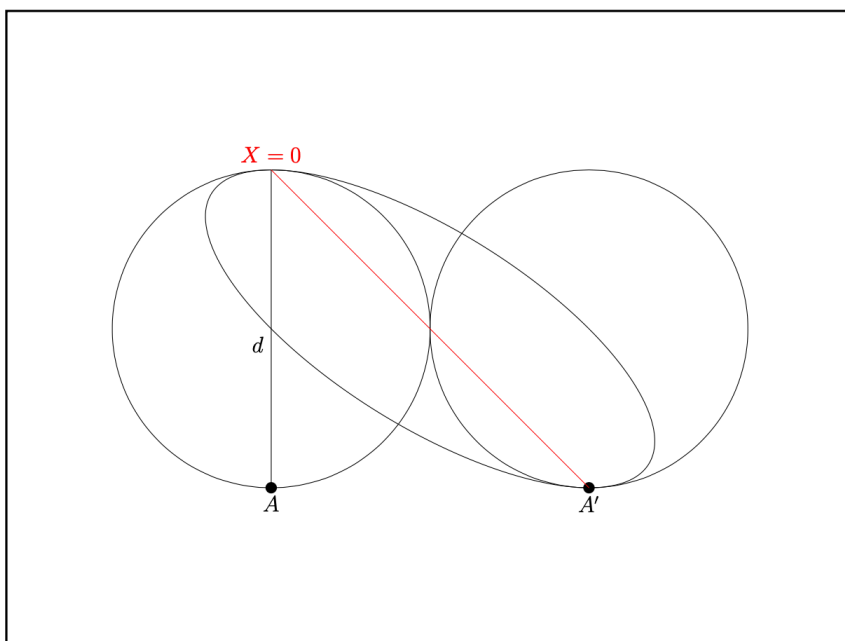
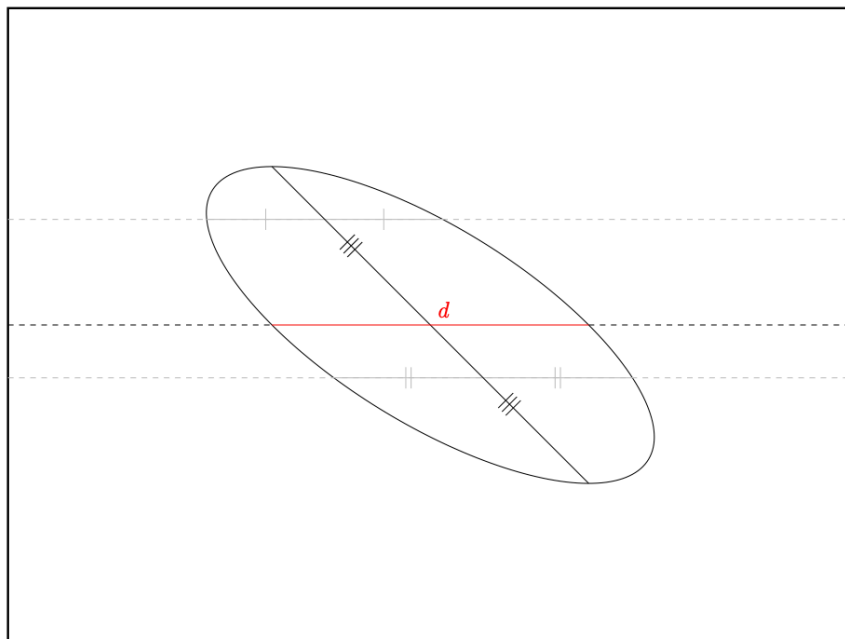
Now we get a segment of length equal to the diameter. To do this, it is enough to draw a horizontal segment through the center of the image $X = 0$.

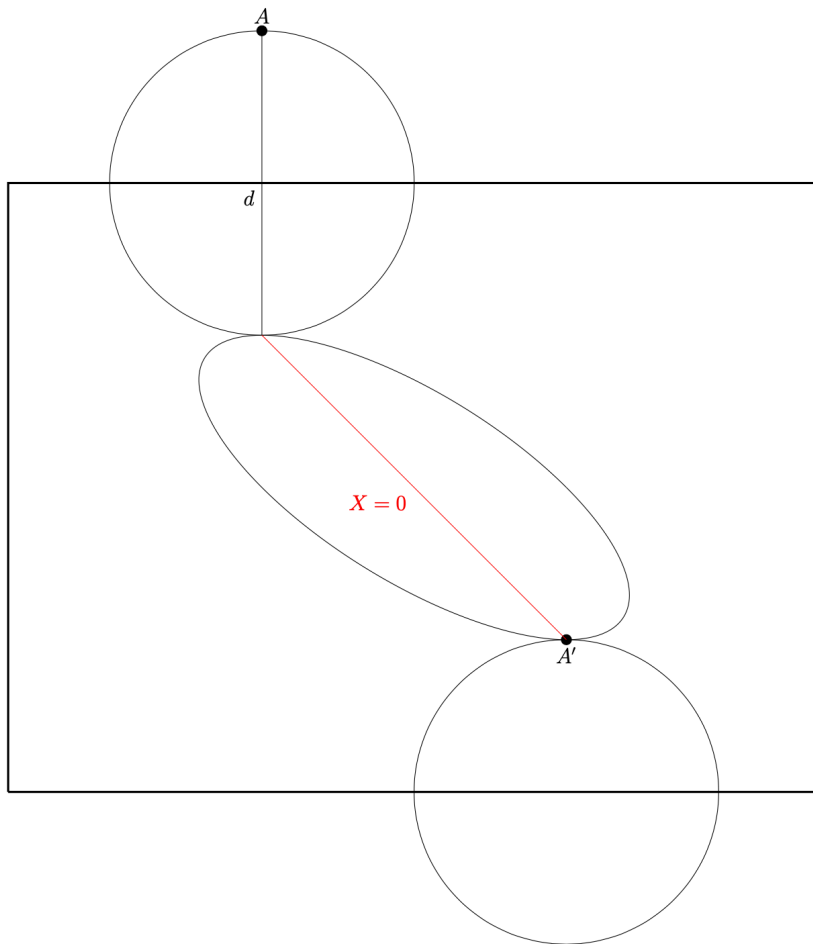
Then there are two options for how photography can take place depending on the speed of the puck. If $u > v \sin \alpha$, then the top point of the image is the top point of the washer. And accordingly, vice versa, if $u < v \sin \alpha$, then the top point of the image is the bottom point of the washer. Let's draw the initial and final states of the puck depending on the speed ratio. $u > v \sin \alpha$:

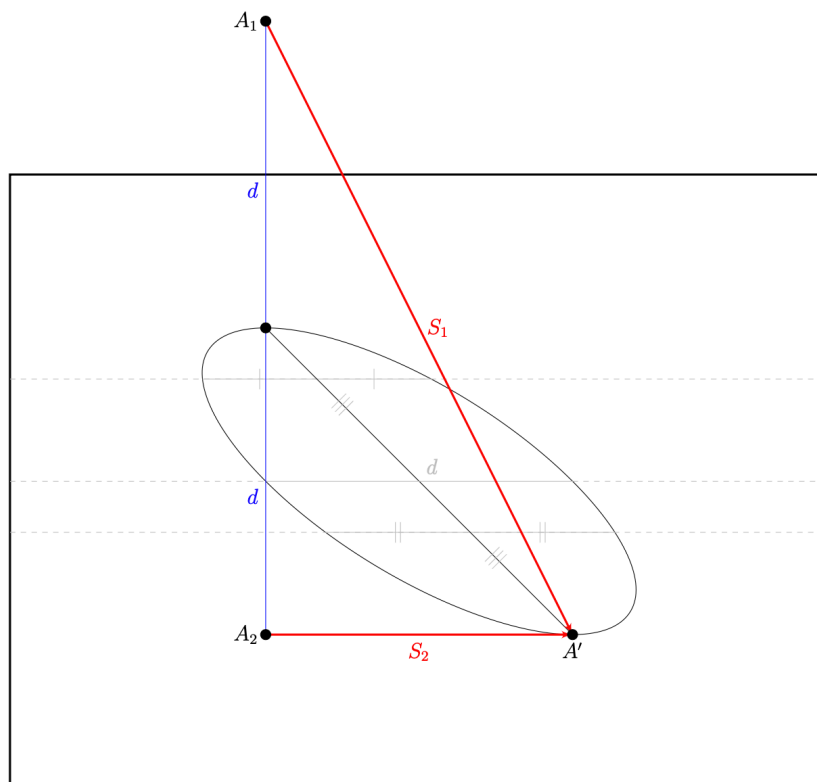
$$u < v \sin \alpha :$$

Note that in this way we can obtain the possible direction of the speed: we obtain the point A by constructing a vertical segment of length d with one end at the top point of the image of the puck, the speed will be directed along $\overrightarrow{AA'}$.









B2

1.00 pts

Using an additional ruler with divisions, find the speed modulus of the puck.

It remains to determine the velocity module. We plotted the displacement of point A during the time of shooting the puck τ . We can express this time in terms of the total photographing time T .

$$\tau = \frac{l}{L}T = 0.5 \cdot T = 5 \text{ ms}$$

По фотографии можем определить displacement шайбы:

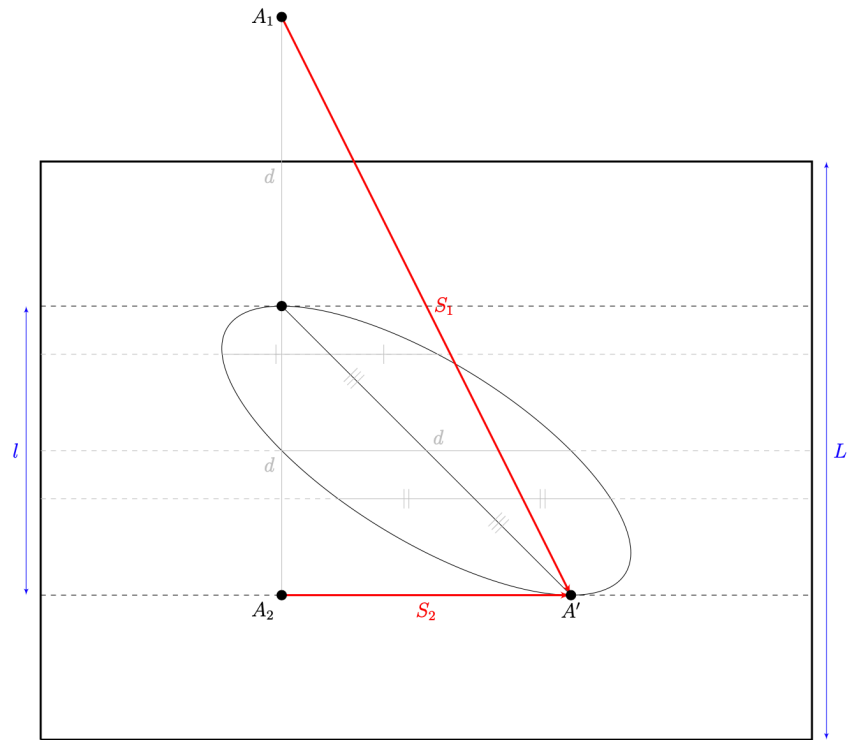
$$\frac{S_1}{d} = 2.24; \quad \frac{S_2}{d} = 1.00.$$

$$S_1 = 17.1 \text{ cm}; \quad S_2 = 7.62 \text{ cm}$$

Осталось посчитать quantity скорости $v = S/\tau$.

Answer: \textbf{Answer:}

$$v_1 = 34.1 \text{ m/s}; \quad v_2 = 15.2 \text{ m/s}$$



C1

0.50 pts

Show that the linear magnification of a pinhole camera is constant and find α .

Let's consider how an image of a single point is formed.

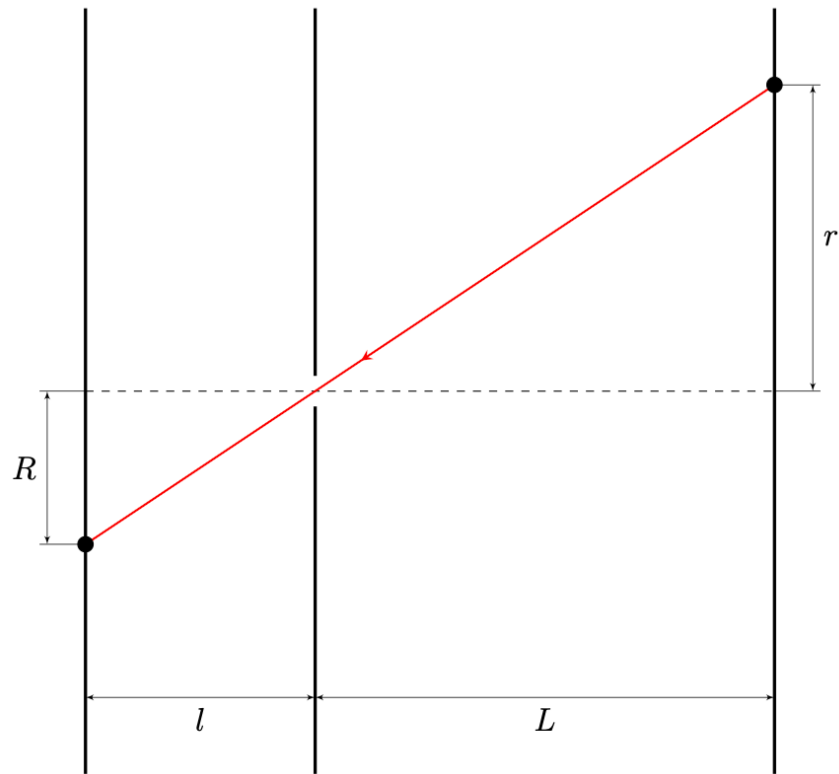
From the similarity we obtain:

$$\frac{R}{r} = \frac{l}{L}.$$

The resulting expression does not depend on the position of the point, so distortion is not observed.

Answer: \textbf{Answer:}

$$\alpha = \frac{l}{L}$$



C2

0.50 pts

Draw a high-quality view of the photograph of part of the notebook sheet shown in the figure below, in cases $\beta > 0$ (positive distortion) and $\beta < 0$ (negative distortion). Consider that $\alpha > 0$.

In the case of $\beta > 0$ the image of the graph paper (we present it for a more detailed illustration) and the square will look like:

In case of $\beta < 0$:

